Evaluating the Spatial and Temporal Patterns of CARE Related Emergency Calls in Seattle

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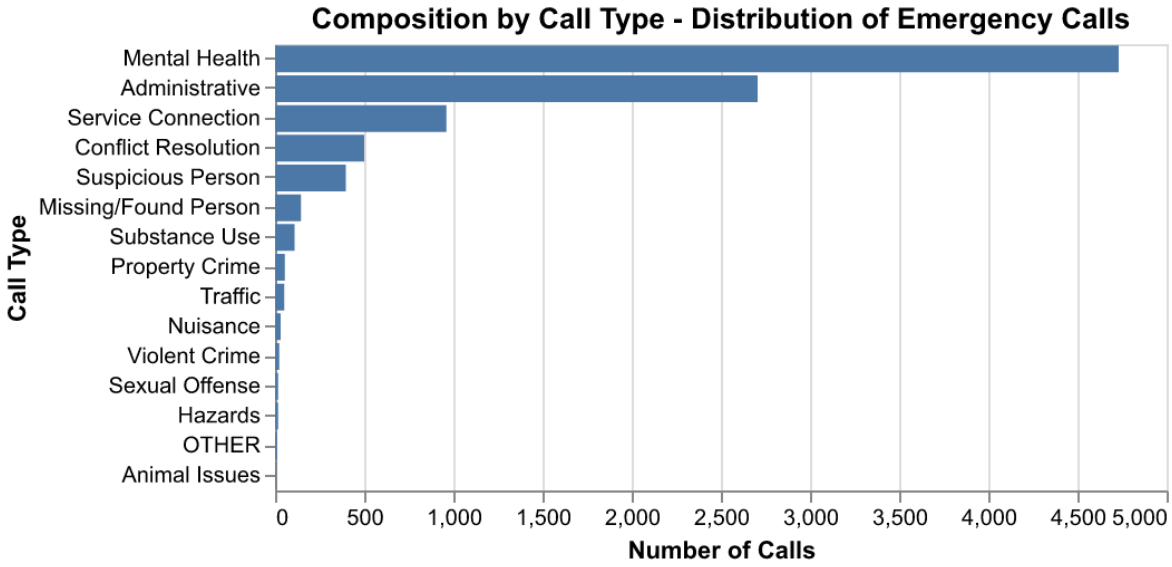
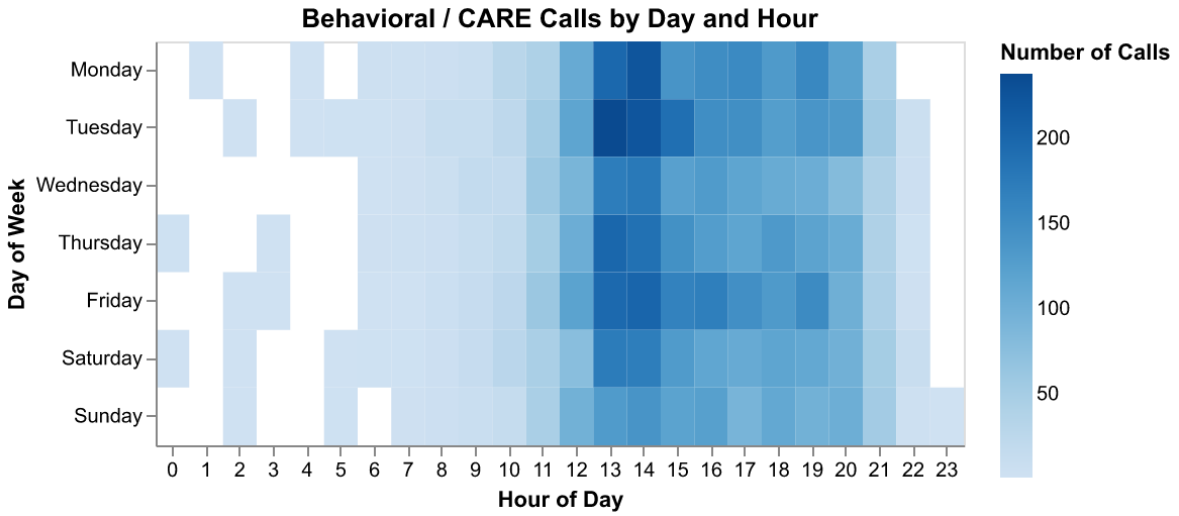
```
/var/folders/js/jqhxbqcd5tn4h65d2pjq_3zm0000gn/T/ipykernel_80769/3877698784.py:33:
UserWarning: Could not infer format, so each element will be parsed
individually, falling back to `dateutil`. To ensure parsing is consistent and
as-expected, please specify a format.
  call_clean[TIME_COL] = pd.to_datetime(call_clean[TIME_COL],
  errors="coerce")

Latitude redacted count: 12017
Longitude redacted count: 12017
```

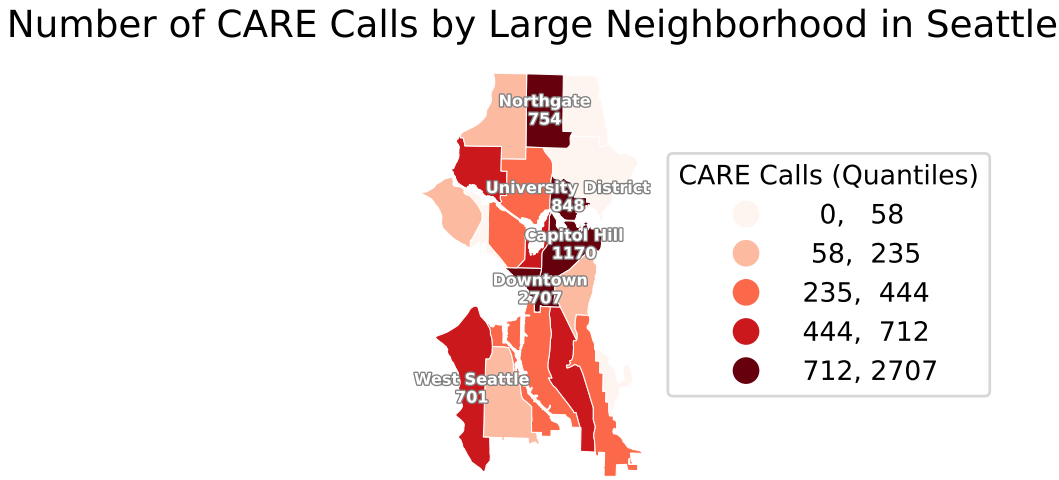


This chart shows the composition of call types within the CARE program specifically. We observe that mental health–related incidents make up the largest share of CARE calls, by

a significant margin. This confirms that CARE is primarily responding to behavioral and psychological crises, which aligns with its intended mission.



This heatmap shows the distribution of behavioral and CARE-related calls by day of the week and hour of the day. We observe a clear temporal pattern: call volumes are relatively low during early morning hours, gradually increase throughout the day, and peak in the early afternoon, particularly between 12 PM and 3 PM. Evening hours remain moderately high before declining after 9 PM. This pattern has important implications for staffing allocation and shift scheduling.



This map shows the number of CARE calls aggregated by large neighborhood in Seattle. We observe a strong concentration of calls in central areas, particularly Downtown, Capitol Hill, and the University District. Downtown stands out with the highest volume, exceeding 2,700

calls. Other high-demand areas include Northgate and West Seattle. This pattern indicates that CARE demand is heavily concentrated in dense, high-activity neighborhoods rather than being evenly distributed across the city.

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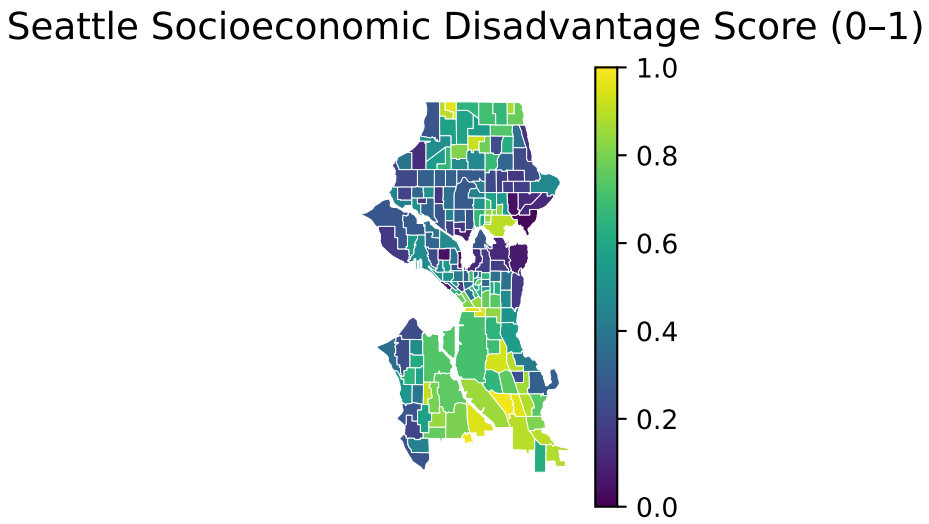
Score range: 0.014 0.9825
```

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Saved to:
/Users/hazellou/student30538/dapII-final-project-nan-sihan-zheng/code/../data/derived-data/ec
```

When we compare the CARE call distribution map with the socioeconomic disadvantage map, an important pattern becomes visible. The highest concentration of CARE calls appears in central neighborhoods such as Downtown and Capitol Hill. However, the highest levels of socioeconomic disadvantage are more concentrated in the southern parts of Seattle. This suggests that high call volume does not perfectly align with areas of greatest socioeconomic vulnerability. From a policy perspective, this highlights the importance of balancing efficiency — responding where calls are highest — with equity — ensuring adequate support in more disadvantaged communities.